

机器人学

第八讲 动力学

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运动规划与误差标定补偿

- 上次课内容提要
 - 关节空间运动规划
 - 末端笛卡尔空间运动规划
 - 误差标定与补偿

机器人的轨迹规划

关节轨迹的插值计算

给定关节空间的起始角度和目标角度，通过插值计算中间时刻的关节角度。

- 三次多项式插值
- 高阶多项式插值
- 用抛物线过渡的线性插值
- B样条插值

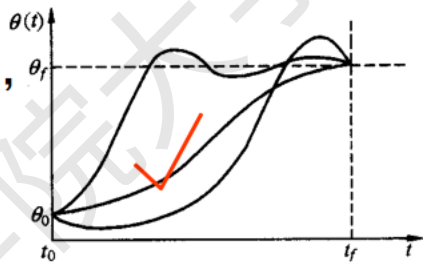
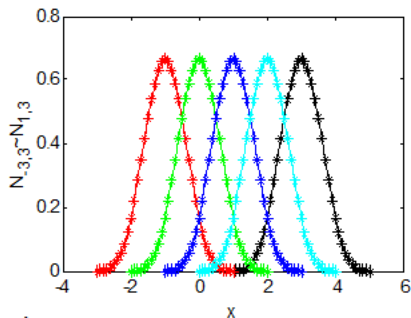


图 7.33 单个关节的不同轨迹曲线

$$N_{i,3}(x) = \begin{cases} \frac{(x-x_i)^3}{(x_{i+1}-x_i)(x_{i+2}-x_i)(x_{i+3}-x_i)}, & x \in [x_i, x_{i+1}) \\ \frac{(x-x_i)^2(x_{i+2}-x)}{(x_{i+2}-x_i)(x_{i+2}-x_{i+1})(x_{i+3}-x_i)} + \frac{(x-x_i)(x-x_{i+1})(x_{i+3}-x)}{(x_{i+2}-x_i)(x_{i+2}-x_{i+1})(x_{i+3}-x_i)}, & x \in [x_{i+1}, x_{i+2}) \\ \frac{(x-x_{i+1})^2(x_{i+4}-x)}{(x_{i+2}-x_{i+1})(x_{i+3}-x_{i+1})(x_{i+4}-x_{i+1})} + \frac{(x-x_i)(x_{i+3}-x)^2}{(x_{i+3}-x_i)(x_{i+3}-x_{i+1})(x_{i+3}-x_{i+2})} + \frac{(x-x_{i+1})(x_{i+3}-x)(x_{i+4}-x)}{(x_{i+3}-x_i)(x_{i+3}-x_{i+1})(x_{i+3}-x_{i+2})}, & x \in [x_{i+2}, x_{i+3}) \\ \frac{(x-x_{i+2})(x_{i+4}-x)^2}{(x_{i+3}-x_{i+2})(x_{i+4}-x_{i+1})(x_{i+4}-x_{i+2})} + \frac{(x_{i+4}-x)^3}{(x_{i+4}-x_{i+1})(x_{i+4}-x_{i+2})(x_{i+4}-x_{i+3})}, & x \in [x_{i+3}, x_{i+4}) \end{cases}$$



$$f(x) = \sum_{i=-m}^k a_i N_{i,m}(x) \quad [x_0, x_{k+1}] \text{ 区间}$$

机器人的轨迹规划

- 笛卡儿空间的插值计算

- 末端运动轨迹为直线：沿空间直线的平移和绕空间转轴的旋转的组合。

假设末端的起始位姿为 $T_1 = \begin{bmatrix} R_1 & P_1 \\ 0 & 1 \end{bmatrix}$ 末端的终止位姿为 $T_2 = \begin{bmatrix} R_2 & P_2 \\ 0 & 1 \end{bmatrix}$

- ❖ 平移矢量，第*i*步的平移矢量

- ❖ 旋转变换矩阵，等效转角与转轴，第*i*步的旋转变换矩阵

- ❖ 第*i*步的位姿

平移矢量为： $P = P_2 - P_1$

第*i*步的平移矢量为： $P(i) = \alpha_i P$

旋转变换矩阵为：
$$R = R_1^T R_2 = \begin{bmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{bmatrix}$$

- 末端运动轨迹为圆弧

- ✓ 求圆心

- ✓ 求半径

- ✓ 求夹角

- ✓ 求第*i*步的位置矢量

机器人的轨迹规划

► Bezier曲线

Bezier曲线的形状能够由少数的几个控制点完全决定，其定义如下：

$$P(t) = \sum_{i=0}^n B_{i,n}(t) Q_i \quad B_{i,n}(t) = C_n^i t^i (1-t)^{n-i} = \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i}$$

$Q_i(x_i, y_i, z_i)$ 是该n次Bezier曲线的n+1个控制点， $i=0, 1, \dots, n$ 。

$$\begin{bmatrix} p_x(t) \\ p_y(t) \\ p_z(t) \end{bmatrix} = \sum_{i=0}^n \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i} \begin{bmatrix} q_{xi} \\ q_{yi} \\ q_{zi} \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^n \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i} q_{xi} \\ \sum_{i=0}^n \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i} q_{yi} \\ \sum_{i=0}^n \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i} q_{zi} \end{bmatrix}$$

机器人的轨迹规划

➤ 人工势场法

- 在机器人运动中，目标对其产生引力场，障碍物对其产生斥力场，则引力场斥力场的叠加称为人工势场

人工势场总势能

$$U_t(q) = U_a(q) + U_r(q)$$

合力

$$f_t(q) = f_a(q) + \sum_{i=1}^p f_{r,i}(q)$$

$$U_a = -\frac{1}{2} k_a \|q - q_g\|^2$$

$$f_a = k_a (q - q_g)$$

$$U_a = k_a \|q - q_g\|$$

$$f_a = k_a \frac{q - q_g}{\|q - q_g\|}$$

$$U_{r,i}(q) = \begin{cases} \frac{k_{r,i}}{\gamma} \left(\frac{1}{\eta_i(q)} - \frac{1}{\eta_{0,i}} \right)^\gamma & \eta_i(q) \leq \eta_{0,i} \\ 0 & \eta_i(q) > \eta_{0,i} \end{cases}$$

$$f_{r,i} = \begin{cases} \frac{k_{r,i}}{\eta_i^2(q)} \left(\frac{1}{\eta_i(q)} - \frac{1}{\eta_{0,i}} \right)^{\gamma-1} \nabla \eta_i(q) & \eta_i(q) \leq \eta_{0,i} \\ 0 & \eta_i(q) > \eta_{0,i} \end{cases}$$

误差标定与补偿

运动参数误差标定

依据D-H方法构建的串联机器人模型的参数误差分为关节变量误差和固定参数误差。为对四参数微小误差进行建模，hayati等人增加了新的参数，引入附加转动变换项 $Rot(y, \beta)$

$$\begin{aligned} T_i &= Trans(a_{i-1}, 0, 0) Rot(x, \alpha_{i-1}) Trans(0, 0, d_i) Rot(z, \theta_i) \\ &= \begin{bmatrix} 1 & 0 & 0 & a_{i-1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha_{i-1} & -\sin \alpha_{i-1} \\ 0 & \sin \alpha_{i-1} & \cos \alpha_{i-1} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & 0 \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$



$$T_i = Trans(a_{i-1}, 0, 0) Rot(x, \alpha_{i-1}) Trans(0, 0, d_i) Rot(z, \theta_i) Rot(y, \beta_i)$$

误差标定与补偿

$$T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_{i-1} \\ \sin\theta_i\cos\alpha_{i-1} & \cos\theta_i\cos\alpha_{i-1} & -\sin\alpha_{i-1} & -d_i\sin\alpha_{i-1} \\ \sin\theta_i\sin\alpha_{i-1} & \cos\theta_i\sin\alpha_{i-1} & \cos\alpha_{i-1} & d_i\cos\alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\beta_i & 0 & \sin\beta_i & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\beta_i & 0 & \cos\beta_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_i\cos\beta_i & -\sin\theta_i & \cos\theta_i\sin\beta_i & a_{i-1} \\ \sin\theta_i\cos\alpha_{i-1}\cos\beta_i + \sin\alpha_{i-1}\sin\beta_i & \cos\theta_i\cos\alpha_{i-1} & \sin\theta_i\cos\alpha_{i-1}\sin\beta_i - \sin\alpha_{i-1}\cos\beta_i & -d_i\sin\alpha_{i-1} \\ \sin\theta_i\sin\alpha_{i-1}\cos\beta_i - \cos\alpha_{i-1}\sin\beta_i & \cos\theta_i\sin\alpha_{i-1} & \sin\theta_i\sin\alpha_{i-1}\sin\beta_i + \cos\alpha_{i-1}\cos\beta_i & d_i\cos\alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

利用微分变换计算微分平移和微分旋转，计算机械臂末端在操作空间的位置误差和方位误差

误差标定与补偿

$${}^T d = M_{\theta} \Delta \theta + M_d \Delta d + M_a \Delta a + M_{\alpha} \Delta \alpha + M_{\beta} \Delta \beta$$

$${}^T \mathcal{S} = R_{\theta} \Delta \theta + R_{\alpha} \Delta \alpha + R_{\beta} \Delta \beta$$

连杆参数误差矢量

$$\Delta \theta = [\Delta \theta_1 \quad \Delta \theta_2 \quad \dots \quad \Delta \theta_n]^T \quad \Delta d = [\Delta d_1 \quad \Delta d_2 \quad \dots \quad \Delta d_n]^T$$

$$\Delta a = [\Delta a_1 \quad \Delta a_2 \quad \dots \quad \Delta a_n]^T \quad \Delta \alpha = [\Delta \alpha_1 \quad \Delta \alpha_2 \quad \dots \quad \Delta \alpha_n]^T$$

$$\Delta \beta = [\Delta \beta_1 \quad \Delta \beta_2 \quad \dots \quad \Delta \beta_n]^T$$

$M_{\theta}, M_d, M_a, M_{\alpha}, M_{\beta}$ $3 \times n$ 偏导数矩阵,
末端位置对运动误差参数求偏导

$R_{\theta}, R_{\alpha}, R_{\beta}$ $3 \times n$ 偏导数矩阵,
末端姿态对运动误差参数求偏导

以上偏导数矩阵可用类似雅可比矩阵的计算方法计算

误差标定与补偿

- 可以观测参考点位置或位姿，建立观测方程 $Bx = b$
- 若B的列矢量线性独立 $B^T Bx = B^T b$
- x 的最小二乘解为 $\hat{x}_{ls} = (B^T B)^{-1} B^T b$

连杆运动参数误差标定算法如下：

步骤一：初始化，将连杆参数调至名义值；

步骤二：计算矩阵M和R，根据名义值和测量数据构建观测方程；

步骤三：最小二乘求解运动参数误差的近似解；

步骤四：根据连杆误差矢量 x 的各分量修改连杆参数值，如 $\theta + \Delta\theta$ ；

步骤五：返回步骤二，直到所得连杆误差矢量 x 的各分量都小于某一最小值为止；

步骤六：连杆运动参数误差为其初始名义值与最后所得到值之差。 10/60

误差标定与补偿

□ 运动参数误差变换补偿

- 基于牛顿-拉弗森的迭代法如下：

步骤一：利用名义关节解，估计预期位置（ p^d ）对应的关节变量；

步骤二：考虑除关节变量误差之外所有其他运动误差（ p^c ），计算正确位置（ p^c ）和偏导数 $\partial p^c / \partial \theta_i (i = 1, 2, \dots, n)$

步骤三：根据方程，利用6个独立分量解出 $\Delta\theta_1, \Delta\theta_2, \dots, \Delta\theta_6$ ，再把所得解与步骤一所得的关节解相加

$$p^d - p^c = \frac{\partial p^c}{\partial \theta_1} \Delta\theta_1 + \frac{\partial p^c}{\partial \theta_2} \Delta\theta_2 + \dots + \frac{\partial p^c}{\partial \theta_6} \Delta\theta_6$$

步骤四：是关节角为步骤三所得值与标定所得的关节误差之差。

机器人动力学

- 本次课内容提要
 - 刚体动力学
 - 机械手动力学

1 刚体动力学

• 基础知识

➤ 拉格朗日(Lagrangian)函数: $L=K-P$, K 为动能, P 为势能

➤ 系统动力学方程: $F_i = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}$, $i=1,2,\dots,n$ 拉格朗日法

F_i 是力或力矩, $\dot{q}_i$ 为速度, q_i 为角位移或线位移

$$\frac{d}{dt} \left(\frac{\partial K}{\partial \dot{q}_i} \right) + \frac{\partial D}{\partial \dot{q}_i} - \left(\frac{\partial K}{\partial q_i} + \frac{\partial P}{\partial q_i} \right) = \frac{\partial W}{\partial q_i} \quad \text{牛顿-欧拉法(Newton-Euler)}$$

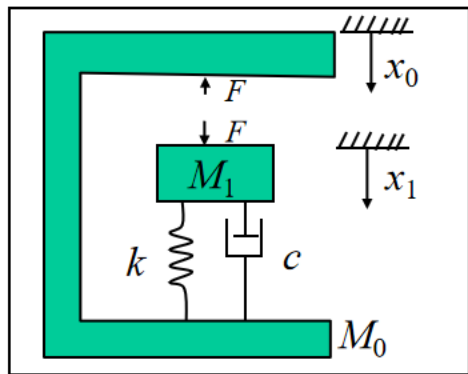
➤ 刚体的动能与位能 (平移式运动)

动能 $K = \frac{1}{2} M_1 \dot{x}_1^2 + \frac{1}{2} M_0 \dot{x}_0^2$

位能 $P = \frac{1}{2} k (x_1 - x_0)^2 - M_1 g x_1 - M_0 g x_0$

耗能 $D = \frac{1}{2} c (\dot{x}_1 - \dot{x}_0)^2$

功 $W = F x_1 - F x_0$



1 刚体动力学

$$\begin{aligned}K &= \frac{1}{2} M_1 \dot{x}_1^2 + \frac{1}{2} M_0 \dot{x}_0^2 \\P &= \frac{1}{2} k (x_1 - x_0)^2 - M_1 g x_1 - M_0 g x_0 \\D &= \frac{1}{2} c (\dot{x}_1 - \dot{x}_0)^2 \\W &= F x_1\end{aligned}$$

► 刚体的动能与位能(平移式运动):

✓ 当 $x_0=0$ 时, x_1 为广义坐标:

$$\frac{d}{dt} \left(\frac{\partial K}{\partial \dot{x}_1} \right) - \frac{\partial K}{\partial x_1} + \frac{\partial D}{\partial \dot{x}_1} + \frac{\partial P}{\partial x_1} = \frac{\partial W}{\partial x_1}$$

牛顿-欧拉法(Newton-Euler)

$$\frac{d}{dt} (M_1 \dot{x}_1) - 0 + c \dot{x}_1 + k x_1 - M_1 g = F \Rightarrow M_1 \ddot{x}_1 + c \dot{x}_1 + k x_1 = F + M_1 g$$

✓ x_0 和 x_1 为广义坐标:

$$\begin{cases} M_1 \ddot{x}_1 + c(\dot{x}_1 - \dot{x}_0) + k(x_1 - x_0) - M_1 g = F \\ M_0 \ddot{x}_0 - c(\dot{x}_1 - \dot{x}_0) - k(x_1 - x_0) - M_0 g = -F \end{cases} \Rightarrow$$

$$\begin{bmatrix} M_1 & 0 \\ 0 & M_0 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_0 \end{bmatrix} + \begin{bmatrix} c & -c \\ -c & c \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_0 \end{bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} x_1 \\ x_0 \end{bmatrix} - \begin{bmatrix} M_1 g \\ M_0 g \end{bmatrix} = \begin{bmatrix} F \\ -F \end{bmatrix}$$

1 刚体动力学

► 刚体的动能与位能（旋转式运动）

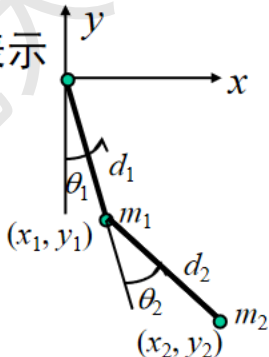
假设连杆质量用等效连杆末端的点质量表示

连杆1:

$$\begin{cases} K_1 = \frac{1}{2} m_1 d_1^2 \dot{\theta}_1^2 \\ P_1 = -m_1 g d_1 \cos \theta_1 \end{cases}$$

连杆2:

$$\begin{cases} K_2 = \frac{1}{2} m_2 v_2^2 \\ P_2 = m_2 g y_2 \end{cases}$$



$$\begin{cases} x_2 = d_1 \sin \theta_1 + d_2 \sin(\theta_1 + \theta_2) \\ y_2 = -d_1 \cos \theta_1 - d_2 \cos(\theta_1 + \theta_2) \end{cases} \Rightarrow \begin{cases} \dot{x}_2 = d_1 \cos \theta_1 \dot{\theta}_1 + d_2 \cos(\theta_1 + \theta_2)(\dot{\theta}_1 + \dot{\theta}_2) \\ \dot{y}_2 = d_1 \sin \theta_1 \dot{\theta}_1 + d_2 \sin(\theta_1 + \theta_2)(\dot{\theta}_1 + \dot{\theta}_2) \end{cases}$$

$$v_2^2 = \dot{x}_2^2 + \dot{y}_2^2 \Rightarrow v_2^2 = d_1^2 \dot{\theta}_1^2 + d_2^2 (\dot{\theta}_1^2 + 2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) + 2d_1 d_2 \cos \theta_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2)$$

$$\begin{cases} K_2 = \frac{1}{2} m_2 d_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 d_2^2 (\dot{\theta}_1^2 + 2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) + m_2 d_1 d_2 \cos \theta_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2) \\ P_2 = -m_2 g d_1 \cos \theta_1 - m_2 g d_2 \cos(\theta_1 + \theta_2) \end{cases}$$

1 刚体动力学

► 刚体的动能与位能（旋转式运动）

$$\begin{cases} K = K_1 + K_2 = \frac{1}{2}(m_1 + m_2)d_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2d_2^2(\dot{\theta}_1^2 + 2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2) + m_2d_1d_2\cos\theta_2(\dot{\theta}_1^2 + \dot{\theta}_1\dot{\theta}_2) \\ P = P_1 + P_2 = -(m_1 + m_2)gd_1\cos\theta_1 - m_2gd_2\cos(\theta_1 + \theta_2) \end{cases}$$

✓ 拉格朗日法求解动力学方程：构造拉格朗日函数 $L=K-P$

$$L = K - P = \frac{1}{2}(m_1 + m_2)d_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2d_2^2(\dot{\theta}_1^2 + 2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2) + m_2d_1d_2\cos\theta_2(\dot{\theta}_1^2 + \dot{\theta}_1\dot{\theta}_2) + (m_1 + m_2)gd_1\cos\theta_1 + m_2gd_2\cos(\theta_1 + \theta_2)$$

$$F_i = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}, \quad i = 1, 2, \dots, n$$

$$\text{求取 } \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1}, \quad \frac{\partial L}{\partial \theta_1}, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_2}, \quad \frac{\partial L}{\partial \theta_2}$$

1 刚体动力学



✓ 拉格朗日法求解动力学方程 (续)

$$\tau_1 = \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} - \frac{\partial L}{\partial \theta_1} = [(m_1 + m_2)d_1^2 + m_2d_2^2 + 2m_2d_1d_2 \cos\theta_2] \ddot{\theta}_1 + (m_2d_2^2 + m_2d_1d_2 \cos\theta_2) \ddot{\theta}_2$$

$$- 2m_2d_1d_2 \sin\theta_2 \dot{\theta}_1 \dot{\theta}_2 - m_2d_1d_2 \sin\theta_2 \dot{\theta}_2^2 + (m_1 + m_2)gd_1 \sin\theta_1 + m_2gd_2 \sin(\theta_1 + \theta_2)$$

$$\tau_2 = \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_2} - \frac{\partial L}{\partial \theta_2} = (m_2d_2^2 + m_2d_1d_2 \cos\theta_2) \ddot{\theta}_1 + m_2d_2^2 \ddot{\theta}_2 + m_2d_1d_2 \sin\theta_2 \dot{\theta}_1^2$$

$$+ m_2gd_2 \sin(\theta_1 + \theta_2)$$

$$\begin{cases} \tau_1 = D_{11} \ddot{\theta}_1 + D_{12} \ddot{\theta}_2 + D_{111} \dot{\theta}_1^2 + D_{122} \dot{\theta}_2^2 + D_{112} \dot{\theta}_1 \dot{\theta}_2 + D_{121} \dot{\theta}_2 \dot{\theta}_1 + D_1 \\ \tau_2 = D_{21} \ddot{\theta}_1 + D_{22} \ddot{\theta}_2 + D_{211} \dot{\theta}_1^2 + D_{222} \dot{\theta}_2^2 + D_{212} \dot{\theta}_1 \dot{\theta}_2 + D_{221} \dot{\theta}_2 \dot{\theta}_1 + D_2 \end{cases}$$



$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} D_{111} & D_{122} \\ D_{211} & D_{222} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1^2 \\ \dot{\theta}_2^2 \end{bmatrix} + \begin{bmatrix} D_{112} & D_{121} \\ D_{212} & D_{221} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \dot{\theta}_2 \\ \dot{\theta}_2 \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} D_1 \\ D_2 \end{bmatrix}$$

力矩

惯量

向心加速度系数

哥氏加速度系数

重力

Coriolis acceleration

1 刚体动力学

✓ 拉格朗日法求解动力学方程（续）

有效惯量：

$$\begin{cases} D_{11} = (m_1 + m_2)d_1^2 + m_2d_2^2 + 2m_2d_1d_2 \cos\theta_2 \\ D_{22} = m_2d_2^2 \end{cases}$$

耦合惯量：

$$D_{12} = D_{21} = m_2d_1d_2 \cos\theta_2$$

向心加速度系数：

$$\begin{cases} D_{111} = 0 \\ D_{122} = -m_2d_1d_2 \sin\theta_2 \\ D_{211} = m_2d_1d_2 \sin\theta_2 \\ D_{222} = 0 \end{cases}$$

哥氏加速度系数：

$$\begin{cases} D_{112} = D_{121} = -m_2d_1d_2 \sin\theta_2 \\ D_{212} = D_{221} = 0 \end{cases}$$

重力项：

$$\begin{cases} D_1 = (m_1 + m_2)gd_1 \sin\theta_1 + m_2gd_2 \sin(\theta_1 + \theta_2) \\ D_2 = m_2gd_2 \sin(\theta_1 + \theta_2) \end{cases}$$



1 刚体动力学

✓ 牛顿-欧拉法求解动力学方程

首先求取动能 K , 位能 P , 消耗能 D , 外力作的功 W

❖ 动能 K

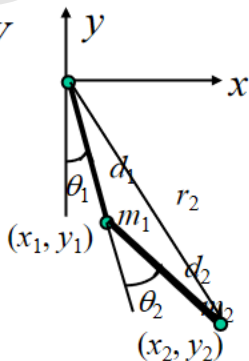
$$\begin{cases} \vec{r}_1 = \vec{r}_0 + d_1 \cos \theta_1 \vec{j} + d_1 \sin \theta_1 \vec{i} = d_1 \cos \theta_1 \vec{j} + d_1 \sin \theta_1 \vec{i} \\ \vec{r}_2 = \vec{r}_1 + d_2 \cos(\theta_1 + \theta_2) \vec{j} + d_2 \sin(\theta_1 + \theta_2) \vec{i} \\ = [d_1 \cos \theta_1 + d_2 \cos(\theta_1 + \theta_2)] \vec{j} + [d_1 \sin \theta_1 + d_2 \sin(\theta_1 + \theta_2)] \vec{i} \end{cases}$$



$$\begin{cases} \vec{v}_1 = \frac{d\vec{r}_1}{dt} = -\dot{\theta}_1 d_1 \sin \theta_1 \vec{j} + \dot{\theta}_1 d_1 \cos \theta_1 \vec{i} \\ \vec{v}_2 = \frac{d\vec{r}_2}{dt} = [-\dot{\theta}_1 d_1 \sin \theta_1 - (\dot{\theta}_1 + \dot{\theta}_2) d_2 \sin(\theta_1 + \theta_2)] \vec{j} + [\dot{\theta}_1 d_1 \cos \theta_1 + (\dot{\theta}_1 + \dot{\theta}_2) d_2 \cos(\theta_1 + \theta_2)] \vec{i} \end{cases}$$



$$\begin{cases} v_1^2 = d_1^2 \dot{\theta}_1^2 \\ v_2^2 = d_1^2 \dot{\theta}_1^2 + d_2^2 (\dot{\theta}_1^2 + 2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) + 2d_1 d_2 \cos \theta_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2) \end{cases}$$



1 刚体动力学

✓ 牛顿-欧拉法求解动力学方程(续)

❖ 动能 K

$$K = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$
$$= \frac{1}{2} (m_1 + m_2) d_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 d_2^2 (\dot{\theta}_1^2 + 2 \dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) + m_2 d_1 d_2 \cos \theta_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2)$$

❖ 位能 P

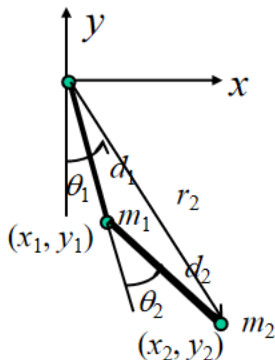
$$P = -m_1 \vec{g} \cdot \vec{r}_1 - m_2 \vec{g} \cdot \vec{r}_2 = -(m_1 + m_2) g d_1 \cos \theta_1 - m_2 g d_2 \cos(\theta_1 + \theta_2)$$

❖ 系统耗能 D

$$D = \frac{1}{2} c_1 \dot{\theta}_1^2 + \frac{1}{2} c_2 \dot{\theta}_2^2$$

❖ 外力做的功 W

$$W = \tau_1 \theta_1 + \tau_2 \theta_2$$



1 刚体动力学

✓ 牛顿-欧拉法求解动力学方程(续)

$$\frac{d}{dt}\left(\frac{\partial K}{\partial \dot{\theta}_i}\right) + \frac{\partial D}{\partial \dot{\theta}_i} - \frac{\partial K}{\partial \theta_i} + \frac{\partial P}{\partial \theta_i} = \frac{\partial W}{\partial \theta_i}$$

$$\begin{aligned}\tau_1 &= [(m_1 + m_2)d_1^2 + m_2 d_2^2 + 2m_2 d_1 d_2 \cos \theta_2] \ddot{\theta}_1 + (m_2 d_2^2 + m_2 d_1 d_2 \cos \theta_2) \ddot{\theta}_2 + c_1 \dot{\theta}_1 \\ &\quad - 2m_2 d_1 d_2 \sin \theta_2 \dot{\theta}_1 \dot{\theta}_2 - m_2 d_1 d_2 \sin \theta_2 \dot{\theta}_2^2 + (m_1 + m_2) g d_1 \sin \theta_1 + m_2 g d_2 \sin(\theta_1 + \theta_2) \\ \tau_2 &= (m_2 d_2^2 + m_2 d_1 d_2 \cos \theta_2) \ddot{\theta}_1 + m_2 d_2^2 \ddot{\theta}_2 + m_2 d_1 d_2 \sin \theta_2 \dot{\theta}_1^2 + c_2 \dot{\theta}_2 + m_2 g d_2 \sin(\theta_1 + \theta_2)\end{aligned}$$

在不考虑消耗的能量时，即 $c_1=c_2=0$ ，则该结果与拉格朗日法的结果相同。

➤ 动力学求解步骤：



- ❖ 求取质点的速度
- ❖ 求质点的动能
- ❖ 求质点的位能
- ❖ 对能量函数求导，获得动力学方程

2 机械手动力学

速度的计算

✓ P点位置

$${}^0\vec{r}_p = T_3 {}^3\vec{r}_p$$

✓ P点速度

$${}^0\vec{v}_p = \frac{d}{dt}({}^0\vec{r}_p) = \dot{T}_3 {}^3\vec{r}_p$$

$$\dot{T}_3 = \sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} \dot{q}_j \Rightarrow {}^0\vec{v}_p = \left(\sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} \dot{q}_j \right) {}^3\vec{r}_p$$

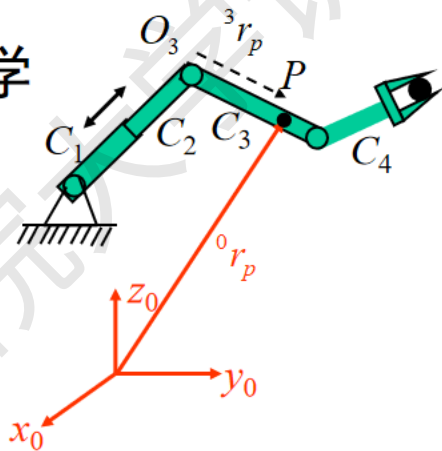
✓ P点加速度

$${}^0\vec{a}_p = \frac{d}{dt}({}^0\vec{v}_p) = \frac{d}{dt} \left(\sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} \dot{q}_j \right) {}^3\vec{r}_p = \left(\sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} \ddot{q}_j \right) {}^3\vec{r}_p + \left(\sum_{k=1}^3 \sum_{j=1}^3 \frac{\partial^2 T_3}{\partial q_k \partial q_j} \dot{q}_k \dot{q}_j \right) {}^3\vec{r}_p$$

✓ P点速度的平方

$$\begin{aligned} ({}^0\mathbf{v}_p)^2 &= {}^0\mathbf{v}_p \cdot {}^0\mathbf{v}_p = \text{Trace} [({}^0\mathbf{v}_p) \cdot ({}^0\mathbf{v}_p)^T] = \text{Trace} \left\{ \left[\left(\sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} \dot{q}_j \right) {}^3\vec{r}_p \right] \cdot \left[\left(\sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} \dot{q}_j \right) {}^3\vec{r}_p \right]^T \right\} \\ &= \text{Trace} \left[\sum_{j=1}^3 \sum_{k=1}^3 \frac{\partial T_3}{\partial q_j} ({}^3\vec{r}_p) ({}^3\vec{r}_p)^T \frac{\partial T_3}{\partial q_k} \dot{q}_j \dot{q}_k \right] \end{aligned}$$

$$\mathbf{v}_p \cdot \mathbf{v}_p^T = \begin{bmatrix} v_{px} \\ v_{py} \\ v_{pz} \end{bmatrix} \begin{bmatrix} v_{px} & v_{py} & v_{pz} \end{bmatrix} = \begin{bmatrix} v_{px}^2 & v_{px}v_{py} & v_{px}v_{pz} \\ v_{py}v_{px} & v_{py}^2 & v_{py}v_{pz} \\ v_{pz}v_{px} & v_{pz}v_{py} & v_{pz}^2 \end{bmatrix}$$



2 机械手动力学

$${}^0\vec{v}_i = {}^0\dot{T}_1 T_2 \cdots {}^{i-1}T_i \vec{r}_i + {}^0T_1 \dot{T}_2 \cdots {}^{i-1}T_i \vec{r}_i + \cdots + {}^0T_1 \cdots {}^{i-1}\dot{T}_i \vec{r}_i + {}^0T_i \dot{\vec{r}}_i = \left(\sum_{j=1}^i \frac{\partial ({}^0T_i)}{\partial q_j} \dot{q}_j \right) {}^i\vec{r}_i$$

对于旋转关节

对于平移关节

$$\begin{aligned} \frac{\partial ({}^{i-1}T_i)}{\partial \theta_i} &= \frac{\partial}{\partial \theta_i} \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -s\theta_i & -c\theta_i c\alpha_i & c\theta_i s\alpha_i & -a_i s\theta_i \\ c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} = Q_i {}^{i-1}T_i \\ \frac{\partial ({}^{i-1}T_i)}{\partial d_i} &= \frac{\partial}{\partial d_i} \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} = Q_i {}^{i-1}T_i \end{aligned}$$

2 机械手动力学

➤ 任意点速度的计算

✓ 连杆*i*上任意一点的位置

$${}^0r = T_i {}^i r$$

✓ 连杆*i*上任意一点的速度

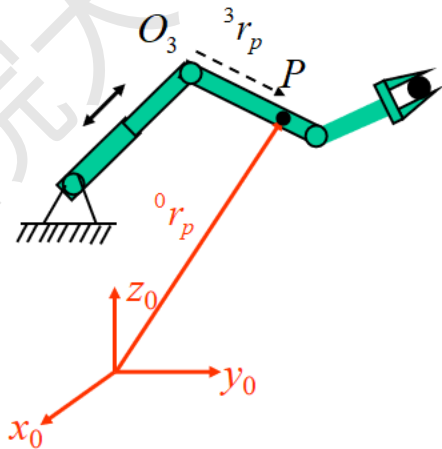
$$v = \left(\sum_{j=1}^i \frac{\partial T_i}{\partial q_j} \dot{q}_j \right) {}^i r$$

✓ 连杆*i*上任意一点的加速度

$$a = \left(\sum_{j=1}^i \frac{\partial T_i}{\partial q_j} \ddot{q}_j \right) {}^i r + \left(\sum_{k=1}^i \sum_{j=1}^i \frac{\partial^2 T_i}{\partial q_k \partial q_j} \dot{q}_k \dot{q}_j \right) {}^i r$$

✓ 连杆*i*上任意一点的速度平方

$$v^2 = \text{Trace} \left[\sum_{k=1}^i \sum_{j=1}^i \frac{\partial T_i}{\partial q_j} {}^i r {}^i r^T \left(\frac{\partial T_i}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right]$$



2 机械手动力学

► 动能 K_i

✓ P 点动能：令连杆3上 P 点的质量为 dm ，则：

$$\begin{aligned} dK_3 &= \frac{1}{2} dm {}^0v_p^2 = \frac{1}{2} dm \text{Trace} \left[\sum_{k=1}^3 \sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} {}^3r_p {}^3r_p^T \left(\frac{\partial T_3}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right] \\ &= \frac{1}{2} \text{Trace} \left[\sum_{k=1}^3 \sum_{j=1}^3 \frac{\partial T_3}{\partial q_j} ({}^3r_p dm {}^3r_p^T) \left(\frac{\partial T_3}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right] \end{aligned}$$

✓ 连杆 i 上任意点的动能

$$\begin{aligned} dK_i &= \frac{1}{2} dm v^2 = \frac{1}{2} dm \text{Trace} \left[\sum_{k=1}^i \sum_{j=1}^i \frac{\partial T_i}{\partial q_j} {}^i r {}^i r^T \left(\frac{\partial T_i}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right] \\ &= \frac{1}{2} \text{Trace} \left[\sum_{k=1}^i \sum_{j=1}^i \frac{\partial T_i}{\partial q_j} ({}^i r dm {}^i r^T) \left(\frac{\partial T_i}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right] \end{aligned}$$

2 机械手动力学

► 动能 K_i

✓ 连杆 i 的动能 $K_i = \int_{\text{连杆 } i} dK_i = \frac{1}{2} \text{Trace} \left[\sum_{k=1}^i \sum_{j=1}^i \frac{\partial T_i}{\partial q_j} \left(\int_{\text{连杆 } i} {}^i \vec{r} {}^i \vec{r}^T dm \right) \left(\frac{\partial T_i}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right]$

$$= \frac{1}{2} \text{Trace} \left[\sum_{k=1}^i \sum_{j=1}^i \frac{\partial T_i}{\partial q_j} I_i \left(\frac{\partial T_i}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right]$$

$$I_i = \int_{\text{连杆 } i} {}^i \vec{r} {}^i \vec{r}^T dm, \quad {}^i \vec{r} {}^i \vec{r}^T = \begin{bmatrix} {}^i x \\ {}^i y \\ {}^i z \\ 1 \end{bmatrix} \begin{bmatrix} {}^i x & {}^i y & {}^i z & 1 \end{bmatrix} = \begin{bmatrix} {}^i x^2 & {}^i x^i y & {}^i x^i z & {}^i x \\ {}^i y^i x & {}^i y^2 & {}^i y^i z & {}^i y \\ {}^i z^i x & {}^i z^i y & {}^i z^2 & {}^i z \\ {}^i x & {}^i y & {}^i z & 1 \end{bmatrix}$$

$$I_i = \int_{\text{连杆 } i} {}^i \vec{r} {}^i \vec{r}^T dm = \begin{bmatrix} \int_{\text{连杆 } i} {}^i x^2 dm & \int_{\text{连杆 } i} {}^i x^i y dm & \int_{\text{连杆 } i} {}^i x^i z dm & \int_{\text{连杆 } i} {}^i x dm \\ \int_{\text{连杆 } i} {}^i y^i x dm & \int_{\text{连杆 } i} {}^i y^2 dm & \int_{\text{连杆 } i} {}^i y^i z dm & \int_{\text{连杆 } i} {}^i y dm \\ \int_{\text{连杆 } i} {}^i z^i x dm & \int_{\text{连杆 } i} {}^i z^i y dm & \int_{\text{连杆 } i} {}^i z^2 dm & \int_{\text{连杆 } i} {}^i z dm \\ \int_{\text{连杆 } i} {}^i x dm & \int_{\text{连杆 } i} {}^i y dm & \int_{\text{连杆 } i} {}^i z dm & \int_{\text{连杆 } i} dm \end{bmatrix}$$

I_i 称为伪转动惯量

2 机械手动力学

► 动能 K_i

✓ 连杆 i 的动能（续）

物体的转动惯量及矩量为：

$$I_{xx} = \int (y^2 + z^2) dm, \quad I_{yy} = \int (x^2 + z^2) dm, \quad I_{zz} = \int (x^2 + y^2) dm$$

$$I_{xy} = I_{yx} = \int xy dm, \quad I_{xz} = I_{zx} = \int xz dm, \quad I_{yz} = I_{zy} = \int yz dm$$

$$mx = \int x dm, \quad my = \int y dm, \quad mz = \int z dm$$

$$\begin{aligned} \int x^2 dm &= -\frac{1}{2} \int (y^2 + z^2) dm + \frac{1}{2} \int (x^2 + z^2) dm + \frac{1}{2} \int (x^2 + y^2) dm \\ &= (-I_{xx} + I_{yy} + I_{zz}) / 2 \end{aligned}$$

$$\begin{aligned} \int y^2 dm &= \frac{1}{2} \int (y^2 + z^2) dm - \frac{1}{2} \int (x^2 + z^2) dm + \frac{1}{2} \int (x^2 + y^2) dm \\ &= (I_{xx} - I_{yy} + I_{zz}) / 2 \end{aligned}$$

$$\begin{aligned} \int z^2 dm &= \frac{1}{2} \int (y^2 + z^2) dm + \frac{1}{2} \int (x^2 + z^2) dm - \frac{1}{2} \int (x^2 + y^2) dm \\ &= (I_{xx} + I_{yy} - I_{zz}) / 2 \end{aligned}$$

2 机械手动力学

➤ 动能 K_t

✓ 连杆 i 的动能 (续)

$$I_i = \begin{bmatrix} (-I_{bxx} + I_{byy} + I_{bzz})/2 & I_{bxy} & I_{bzx} & m_i \bar{x}_i \\ I_{bxy} & (I_{bxx} - I_{byy} + I_{bzz})/2 & I_{byz} & m_i \bar{y}_i \\ I_{bzx} & I_{byz} & (I_{bxx} + I_{byy} - I_{bzz})/2 & m_i \bar{z}_i \\ m_i \bar{x}_i & m_i \bar{y}_i & m_i \bar{z}_i & m_i \end{bmatrix}$$

✓ n 个连杆的动能

$$K = \sum_{i=1}^n K_i = \frac{1}{2} \sum_{i=1}^n \text{Trace} \left[\sum_{k=1}^i \sum_{j=1}^i \frac{\partial T_i}{\partial q_j} I_i \left(\frac{\partial T_i}{\partial q_k} \right)^T \dot{q}_j \dot{q}_k \right] = \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^i \sum_{j=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_j} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_j \dot{q}_k$$

✓ 连杆 i 传动装置的动能

$$K_{ai} = \frac{1}{2} I_{ai} \dot{q}_i^2$$

✓ 所有连杆传动装置的动能

$$K_a = \frac{1}{2} \sum_{i=1}^n I_{ai} \dot{q}_i^2$$

✓ 动能 K_t

$$K_t = K + K_a = \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^i \sum_{j=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_j} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_j \dot{q}_k + \frac{1}{2} \sum_{i=1}^n I_{ai} \dot{q}_i^2$$

2 机械手动力学

➤ 位能 P

- ✓ 连杆 i 的位置 i_r 处的质点 dm 位能

$$dP_i = -dm \bar{g}^T {}^0 \bar{r} = -\bar{g}^T T_i^i \bar{r} dm \quad \bar{g}^T = \begin{bmatrix} g_x & g_y & g_z & 1 \end{bmatrix}$$

- ✓ 连杆 i 的位能

$$P_i = \int_{\text{连杆 } i} dP_i = - \int_{\text{连杆 } i} \bar{g}^T T_i^i \bar{r} dm = -m_i \bar{g}^T T_i^i \bar{r}$$

- ✓ n 个连杆的位能，传动装置的位能 P_{ai} 忽略不记

$$P = \sum_{i=1}^n (P_i - P_{ai}) \approx \sum_{i=1}^n P_i = - \sum_{i=1}^n m_i \bar{g}^T T_i^i \bar{r}$$

➤ 构造拉格朗日函数

$$L = K_t - P = \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^i \sum_{j=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_j} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_j \dot{q}_k + \frac{1}{2} \sum_{i=1}^n I_{ai} \dot{q}_i^2 + \sum_{i=1}^n m_i \bar{g}^T T_i^i \bar{r}$$

2 机械手动力学

➤ 动力学求解

$$\frac{\partial L}{\partial \dot{q}_p} = \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_k + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_j} I_i \frac{\partial T_i^T}{\partial q_p} \right) \dot{q}_j + I_{ap} \dot{q}_p$$

$p = 1, 2, \dots, n$

由于 I_i 为对称矩阵，所以

$$\text{Trace} \left(\frac{\partial T_i}{\partial q_j} I_i \frac{\partial T_i^T}{\partial q_k} \right) = \text{Trace} \left(\frac{\partial T_i}{\partial q_k} I_i^T \frac{\partial T_i^T}{\partial q_j} \right) = \text{Trace} \left(\frac{\partial T_i}{\partial q_k} I_i \frac{\partial T_i^T}{\partial q_j} \right) \rightarrow$$

$$\begin{aligned} \frac{\partial L}{\partial \dot{q}_p} &= \sum_{i=1}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_k + I_{ap} \dot{q}_p \\ &= \sum_{i=p}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_k + I_{ap} \dot{q}_p \end{aligned}$$

当 $p > i$ 时，后面连杆对前面无影响(没有对应变量)

2 机械手动力学

► 动力学求解 (续)

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{\partial T_i}{\partial \dot{q}_j} \right) &= \sum_{k=1}^i \frac{\partial}{\partial q_k} \left(\frac{\partial T_i}{\partial \dot{q}_j} \right) \dot{q}_k = \sum_{k=1}^i \frac{\partial^2 T_i}{\partial q_k \partial \dot{q}_j} \dot{q}_k \\
 \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_p} \right) &= \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_j \partial q_k} I_i \frac{\partial T_i^T}{\partial q_p} \right) \dot{q}_j \dot{q}_k + \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_p \partial q_k} I_i \frac{\partial T_i^T}{\partial q_j} \right) \dot{q}_j \dot{q}_k \\
 &\quad + \sum_{i=p}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \ddot{q}_k + I_{ap} \ddot{q}_p \\
 &= \sum_{i=p}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \ddot{q}_k + I_{ap} \ddot{q}_p + 2 \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_j \partial q_k} I_i \frac{\partial T_i^T}{\partial q_p} \right) \dot{q}_j \dot{q}_k \\
 \frac{\partial L}{\partial q_p} &= \frac{1}{2} \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_j \partial q_k} I_i \frac{\partial T_i^T}{\partial q_p} \right) \dot{q}_j \dot{q}_k + \frac{1}{2} \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_p \partial q_k} I_i \frac{\partial T_i^T}{\partial q_j} \right) \dot{q}_j \dot{q}_k \\
 &\quad + \sum_{i=p}^n m_i \bar{\mathbf{g}}^T \frac{\partial T_i}{\partial q_p} \bar{\mathbf{r}}_i \\
 &= \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_p \partial q_k} I_i \frac{\partial T_i^T}{\partial q_j} \right) \dot{q}_j \dot{q}_k + \sum_{i=p}^n m_i \bar{\mathbf{g}}^T \frac{\partial T_i}{\partial q_p} \bar{\mathbf{r}}_i
 \end{aligned}$$

2 机械手动力学

► 动力学求解（续）

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_p} \right) - \frac{\partial L}{\partial q_p} &= \sum_{i=p}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \ddot{q}_k + I_{ap} \ddot{q}_p + 2 \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_j \partial q_k} I_i \frac{\partial T_i^T}{\partial q_p} \right) \dot{q}_j \dot{q}_k \\
 &\quad - \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_p \partial q_k} I_i \frac{\partial T_i^T}{\partial q_j} \right) \dot{q}_j \dot{q}_k - \sum_{i=p}^n m_i \bar{g}^T \frac{\partial T_i}{\partial q_p} \bar{r}_i \\
 &= \sum_{i=p}^n \sum_{k=1}^i \text{Trace} \left(\frac{\partial T_i}{\partial q_p} I_i \frac{\partial T_i^T}{\partial q_k} \right) \ddot{q}_k + I_{ap} \ddot{q}_p + \sum_{i=p}^n \sum_{j=1}^i \sum_{k=1}^i \text{Trace} \left(\frac{\partial^2 T_i}{\partial q_j \partial q_k} I_i \frac{\partial T_i^T}{\partial q_p} \right) \dot{q}_j \dot{q}_k - \sum_{i=p}^n m_i \bar{g}^T \frac{\partial T_i}{\partial q_p} \bar{r}_i \\
 \tau_i &= \sum_{j=i}^n \sum_{k=1}^j \text{Trace} \left(\frac{\partial T_j}{\partial q_i} I_j \frac{\partial T_j^T}{\partial q_k} \right) \ddot{q}_k + I_{ai} \ddot{q}_i + \sum_{j=i}^n \sum_{k=1}^j \sum_{m=1}^j \text{Trace} \left(\frac{\partial^2 T_j}{\partial q_k \partial q_m} I_j \frac{\partial T_j^T}{\partial q_i} \right) \dot{q}_k \dot{q}_m - \sum_{j=i}^n m_j \bar{g}^T \frac{\partial T_j}{\partial q_i} \bar{r}_j \\
 \tau_i &= \sum_{j=1}^n D_{ij} \ddot{q}_k + I_{ai} \ddot{q}_i + \sum_{j=i}^n \sum_{k=1}^j D_{ijk} \dot{q}_j \dot{q}_k + D_i \\
 D_{ij} &= \sum_{p=\max i, j}^j \text{Trace} \left(\frac{\partial T_p}{\partial q_j} I_p \frac{\partial T_p^T}{\partial q_i} \right), D_{ijk} = \sum_{p=\max i, j, k}^n \text{Trace} \left(\frac{\partial^2 T_p}{\partial q_k \partial q_m} I_p \frac{\partial T_p^T}{\partial q_i} \right), D_i = - \sum_{p=i}^n m_p \bar{g}^T \frac{\partial T_p}{\partial q_i} \bar{r}_p
 \end{aligned}$$

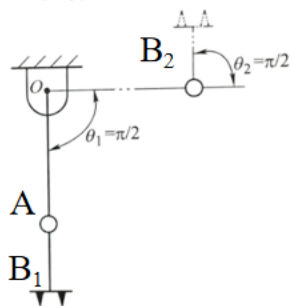
惯量

向心加速度系数 / 哥氏加速度系数

重力

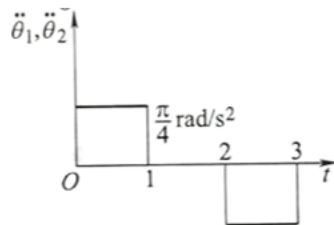
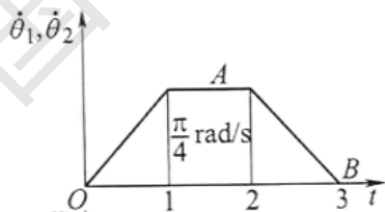
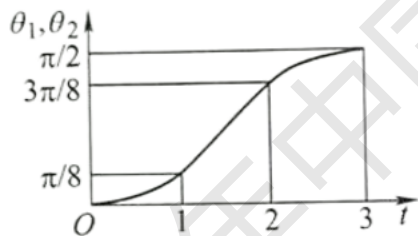
2 机械手动力学

例题



- 如图所示平面机械手，在O点和A点处分别装有驱动电机，产生力矩 τ_1 、 τ_2 带动机械手运动。

$d_1=d_2=1\text{m}$ ，臂杆自重不计，点A处假定集中质量 $m_1=2\text{kg}$ ，B处夹持器和负载合计质量为 $m_2=4\text{kg}$ 。考虑在无重力环境下运动，在3s内从向下位置等加速-等速-等减速各轴分别转过 90° ，将负载由B1移动之B2，计算力矩 τ_1 、 τ_2 并确定运动规律是否可行，如何修改。



2 机械手动力学

$$\tau_1 = [(m_1 + m_2)d_1^2 + m_2d_2^2 + 2m_2d_1d_2 \cos \theta_2]\ddot{\theta}_1 + (m_2d_2^2 + m_2d_1d_2 \cos \theta_2)\ddot{\theta}_2 - 2m_2d_1d_2 \sin \theta_2 \dot{\theta}_1 \dot{\theta}_2 - m_2d_1d_2 \sin \theta_2 \dot{\theta}_2^2$$

$$\tau_2 = (m_2d_2^2 + m_2d_1d_2 \cos \theta_2)\ddot{\theta}_1 + m_2d_2^2\ddot{\theta}_2 + m_2d_1d_2 \sin \theta_2 \dot{\theta}_1^2$$

代入参数

$$\tau_1 = (10 + 8 \cos \theta_2)\ddot{\theta}_1 + (4d_2^2 + 4 \cos \theta_2)\ddot{\theta}_2 - 8 \sin \theta_2 \dot{\theta}_1 \dot{\theta}_2 - 4 \sin \theta_2 \dot{\theta}_2^2$$

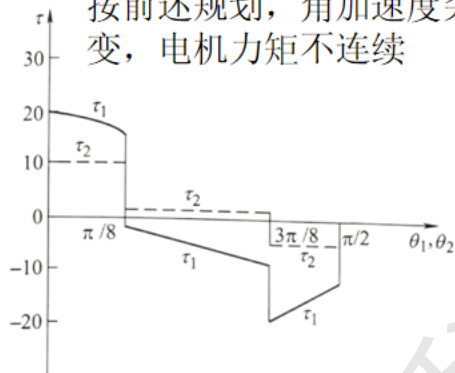
$$\tau_2 = (4 + 4 \cos \theta_2)\ddot{\theta}_1 + 4\ddot{\theta}_2 + 4 \sin \theta_2 \dot{\theta}_1^2$$

轨迹规划的参数

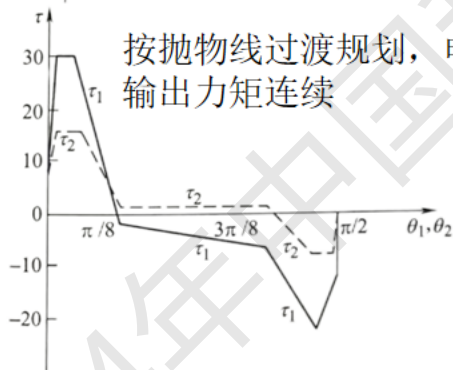
$$\theta_1 = \theta_2 = \begin{cases} \pi t^2 / 8 \\ \pi(-1 + 2t) / 8 \\ \pi(-5 + 6t - t^2) / 8 \end{cases} \quad \dot{\theta}_1 = \dot{\theta}_2 = \begin{cases} \pi t / 4 \\ \pi / 4 \\ \pi(3 - t) / 4 \end{cases} \quad \ddot{\theta}_1 = \ddot{\theta}_2 = \begin{cases} \pi / 4 & 0-1s \\ 0 & 1-2s \\ -\pi / 4 & 2-3s \end{cases}$$

2 机械手动力学

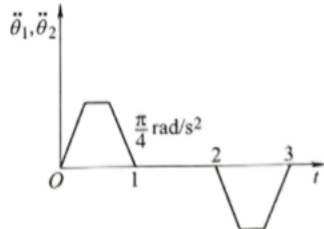
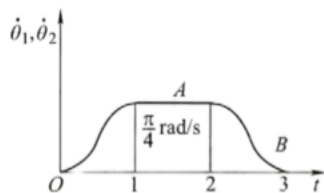
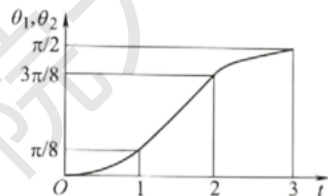
按前述规划，角加速度突变，电机力矩不连续



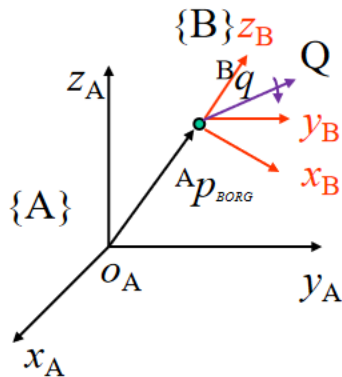
按抛物线过渡规划，电机输出力矩连续



按抛物线过渡轨迹重新规划，确保角加速度连续



2 机械手动力学



{B}系相对于{A}系位置向量为 ${}^A p_{BORG}$ ；{B}系相对于{A}系旋转矩阵 ${}^A R_B$ ；{B}系中向量 ${}^B q$ ，其相对{B}系原点速度为 ${}^B \dot{V}_Q$ ；Q相对于{A}系的线速度为 ${}^A \dot{V}_Q$ ；相对于{A}系的角速度 ${}^A \dot{\Omega}_B$

刚体线速度

$${}^A \dot{V}_Q = {}^A \dot{V}_{BORG} + {}^A R^B \dot{V}_Q + {}^A \dot{\Omega}_B \times ({}^A R^B q)$$

求导得刚体线加速度

$$\begin{aligned} {}^A \ddot{V}_Q = & {}^A \ddot{V}_{BORG} + {}^A \ddot{R}^B \dot{V}_Q + {}^A \dot{R}^B \ddot{V}_Q \\ & + {}^A \dot{\Omega}_B \times ({}^A R^B q) + {}^A \Omega_B \times ({}^A \dot{R}^B q) + {}^A \Omega_B \times ({}^A R^B \dot{q}) \end{aligned}$$

$$\text{因为 } {}^A \dot{R} = {}^A \Omega_B \times {}^A R \quad {}^B \dot{q} = {}^B \dot{V}_Q$$

$$\begin{aligned} {}^A \ddot{V}_Q = & {}^A \ddot{V}_{BORG} + {}^A \Omega_B \times ({}^A R^B \dot{V}_Q) + {}^A \dot{R}^B \ddot{V}_Q \\ & + {}^A \dot{\Omega}_B \times ({}^A R^B q) + {}^A \Omega_B \times ({}^A \Omega_B \times {}^A R^B q) \\ & + {}^A \Omega_B \times ({}^A R^B \dot{V}_Q) \end{aligned}$$

2 机械手动力学

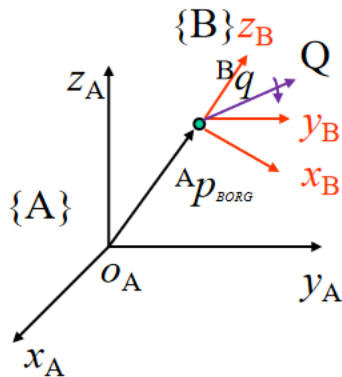
$$\begin{aligned} {}^A\dot{V}_Q &= {}^A\dot{V}_{BORG} + {}^A R^B \dot{V}_Q && \text{线加速度} \\ &+ 2 {}^A \Omega_B \times ({}^A R^B V_Q) && \text{哥氏加速度} \\ &+ {}^A \dot{\Omega}_B \times ({}^A R^B q) && \text{欧拉加速度} \\ &+ {}^A \Omega_B \times ({}^A \Omega_B \times {}^A R^B q) && \text{向心加速度} \end{aligned}$$

但 Bq 为常量时, ${}^B V_Q = {}^B \dot{V}_Q = 0$

线速度可简化为:

$${}^A\dot{V}_Q = {}^A\dot{V}_{BORG} + {}^A \dot{\Omega}_B \times ({}^A R^B q) + {}^A \Omega_B \times ({}^A \Omega_B \times {}^A R^B q)$$

2 机械手动力学



刚体角速度

$${}^A\Omega_C = {}^A\Omega_B + {}^A_R{}^B\Omega_C$$

求导得刚体角加速度

$${}^A\dot{\Omega}_C = {}^A\dot{\Omega}_B + {}^A_R{}^B\dot{\Omega}_C + {}^A\dot{R}{}^B\Omega_C$$

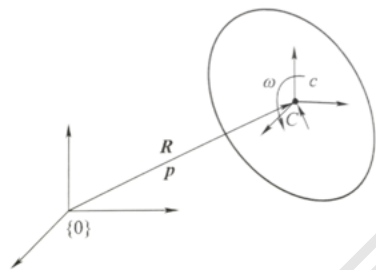
{B}系以角速度 ${}^A\Omega_B$ 相对于
{A}系转动；{C}系以角速
度 ${}^B\Omega_C$ 相对于{B}系转动，
则{C}系相对于{A}系转动
角速度为 ${}^A\Omega_C$

因为 ${}^A\dot{R} = {}^A\Omega_B \times {}^A_R$

$${}^A\dot{\Omega}_C = {}^A\dot{\Omega}_B + {}^A_R{}^B\dot{\Omega}_C + {}^A\Omega_B \times ({}^A_R{}^B\Omega_C)$$

2 机械手动力学

牛顿-欧拉方程



牛顿方程

$$f = \frac{d(mv_c)}{dt} = m\dot{v}_c$$

刚体质量、质心线加速度、质心处合力

欧拉方程

$$\tau = \frac{d({}^c I \omega)}{dt} = {}^c I \dot{\omega} + \omega \times ({}^c I \omega)$$

${}^c I$ 是刚体相对质心坐标系的惯性张量；刚体角速度、角加速度； τ 为质心处合力矩

将欧拉方程转为基坐标系{0}

$${}^0 \tau = {}^0 R {}^c I {}^0 R^T {}^0 \dot{\omega} + {}^0 \omega \times ({}^0 R {}^c I {}^0 R^T) {}^0 \omega$$

$${}^0 \tau = {}^0 I {}^0 \dot{\omega} + {}^0 \omega \times ({}^0 I {}^0 \omega)$$

2 机械手动力学

连杆速度加速度的向外递推公式

转动关节的速度和加速度传递

连杆*i*在其自身{*i*}坐标系下的线速度和角速度为 $({}^i\mathbf{v}_i, {}^i\boldsymbol{\omega}_i)^\top$

$${}^i\boldsymbol{\omega}_{i+1} = {}^i\boldsymbol{\omega}_i + {}^iR_{i+1}\dot{\theta}_{i+1} {}^{i+1}\mathbf{z}_{i+1} \quad {}^{i+1}\mathbf{z}_{i+1} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

${}^{i+1}\boldsymbol{\omega}_{i+1}$ 及加速度相对于{*i+1*}坐标系表示

$${}^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}R_i {}^i\boldsymbol{\omega}_i + \dot{\theta}_{i+1} {}^{i+1}\mathbf{z}_{i+1}$$

$${}^{i+1}\dot{\boldsymbol{\omega}}_{i+1} = {}^{i+1}R_i \dot{{}^i\boldsymbol{\omega}}_i + {}^{i+1}\mathbf{z}_{i+1} \ddot{\theta}_{i+1} + {}^{i+1}R_i {}^i\boldsymbol{\omega}_i \times {}^{i+1}\mathbf{z}_{i+1} \dot{\theta}_{i+1}$$

${}^{i+1}\mathbf{v}_{i+1}$ 相对于{*i+1*}坐标系的表示 ${}^i\mathbf{v}_{i+1} = {}^i\mathbf{v}_i + {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{r}_{i+1} \quad {}^{i+1}\mathbf{v}_{i+1} = {}^{i+1}R_i ({}^i\mathbf{v}_i + {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{r}_{i+1})$

线加速度相对于{*i+1*}坐标系表示 ${}^{i+1}\dot{\mathbf{v}}_{i+1} = {}^{i+1}R_i [{}^i\dot{\mathbf{v}}_i + {}^i\dot{\boldsymbol{\omega}}_i \times {}^i\mathbf{r}_{i+1} + {}^i\boldsymbol{\omega}_i \times ({}^i\boldsymbol{\omega}_i \times {}^i\mathbf{r}_{i+1})]$

2 机械手动力学

连杆速度加速度的向外递推公式

移动关节的速度和加速度传递

连杆*i*在其自身{*i*}坐标系下的线速度和角速度为 $({}^i\omega_i, {}^i\mathbf{v}_i)^\top$

相对于{*i*+1}坐标系线速度和角速度表示

$${}^{i+1}\omega_{i+1} = {}^{i+1}_i R {}^i\omega_i$$

$${}^{i+1}\mathbf{v}_{i+1} = {}^{i+1}_i R ({}^i\mathbf{v}_i + {}^i\omega_i \times {}^i\mathbf{r}_{i+1}) + \dot{d}_{i+1} {}^{i+1}\mathbf{z}_{i+1}$$

相对于{*i*+1}坐标系线加速度和角加速度

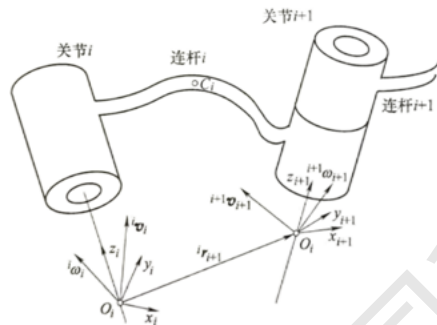
$${}^{i+1}\dot{\omega}_{i+1} = {}^{i+1}_i R {}^i\dot{\omega}_i$$

$${}^{i+1}\dot{\mathbf{v}}_{i+1} = {}^{i+1}_i R \left[{}^i\dot{\mathbf{v}}_i + {}^i\dot{\omega}_i \times {}^i\mathbf{r}_{i+1} + {}^i\omega_i \times ({}^i\omega_i \times {}^i\mathbf{r}_{i+1}) \right] + 2{}^{i+1}\omega_{i+1} \times {}^{i+1}\mathbf{z}_{i+1} \dot{d}_{i+1} + {}^{i+1}\mathbf{z}_{i+1} \ddot{d}_{i+1}$$

2 机械手动力学

连杆速度加速度的向外递推公式

连杆质心的速度和加速度传递



连杆 i 质心坐标系为 $\{C_i\}$ ，固连在连杆 i 上，坐标轴方向与 $\{i\}$ 一致，则其质心速度和加速度

$${}^{i+1}\mathbf{v}_{C_{i+1}} = {}^{i+1}\mathbf{v}_{i+1} + {}^{i+1}\boldsymbol{\omega}_{i+1} \times {}^i\mathbf{r}_{C_{i+1}}$$

$${}^{i+1}\dot{\mathbf{v}}_{C_{i+1}} = {}^{i+1}\dot{\mathbf{v}}_{i+1} + {}^{i+1}\dot{\boldsymbol{\omega}}_{i+1} \times {}^i\mathbf{r}_{C_{i+1}} + {}^{i+1}\boldsymbol{\omega}_{i+1} \times ({}^{i+1}\boldsymbol{\omega}_{i+1} \times {}^i\mathbf{r}_{C_{i+1}})$$

在连杆速度和加速度外推时，递推初始值为

$${}^0\boldsymbol{\omega}_0 = {}^0\mathbf{v}_0 = {}^0\dot{\boldsymbol{\omega}}_0 = {}^0\dot{\mathbf{v}}_0 = 0$$

2 机械手动力学

关节力和力矩的向内递推公式

在无运动，静力/力矩平衡方程

$${}^i f_i - {}^i f_{i+1} + {}^i m_i g = 0$$

$${}^i M_i - {}^i M_{i+1} - {}^i r_{i+1} \times {}^i f_{i+1} + {}^i r_{Ci} \times {}^i m_i g = 0$$

在运动时

$${}^i f_{Ci} = m_i \dot{v}_{Ci} \quad {}^i M_{Ci} = {}^i I_{Ci} \dot{\omega}_i + {}^i \omega_i \times {}^i I_{Ci} {}^i \omega_i$$

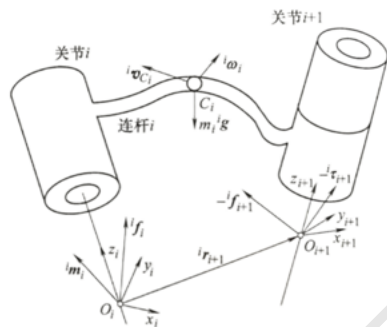
质心处合力为零 ${}^i f_{Ci} = {}^i f_i - {}^i R^{i+1} f_{i+1} \Rightarrow {}^i f_i = {}^i f_{Ci} + {}^i R^{i+1} f_{i+1}$

质心处合力矩为零 ${}^i M_{Ci} = {}^i M_i - {}^i R^{i+1} M_{i+1} - {}^i r_{i+1} \times {}^i R^{i+1} f_{i+1} - {}^i r_{Ci} \times {}^i f_{Ci}$

$\Rightarrow {}^i M_i = {}^i M_{Ci} + {}^i R^{i+1} M_{i+1} + {}^i r_{i+1} \times {}^i R^{i+1} f_{i+1} + {}^i r_{Ci} \times {}^i f_{Ci}$

2 机械手动力学

- 从末端开始，依次向内递推($n \rightarrow 1$)，计算机器人各连杆对相邻连杆施加的力和力矩。各关节所需的转矩等于连杆作用在其邻杆上力矩的Z轴分量。



$$\tau_i = {}^i M_i^T z_i$$

移动关节驱动力

$$\tau_i = {}^i f_i^T z_i$$

在机器人无负载，自由运动时 ${}^{n+1} f_{n+1} = {}^{n+1} M_{n+1} = 0$

其他运动时，末端连杆力和力矩等于其与环境的接触力/力矩

2 机械手动力学

- 计算速度、加速度外推公式 ($i = 0 \rightarrow n$)

$${}^0\omega_0 = {}^0v_0 = {}^0\dot{\omega}_0 = {}^0\dot{v}_0 = 0$$

$${}^{i+1}\omega_{i+1} = \begin{cases} {}^{i+1}R^i \omega_i + \dot{\theta}_{i+1} {}^{i+1}z_{i+1} & \text{转动关节} \\ {}^{i+1}R^i \omega_i & \text{移动关节} \end{cases}$$

$${}^{i+1}\dot{\omega}_{i+1} = \begin{cases} {}^{i+1}R^i \dot{\omega}_i + {}^{i+1}z_{i+1} \ddot{\theta}_{i+1} + {}^{i+1}R^i \omega_i \times {}^{i+1}z_{i+1} \dot{\theta}_{i+1} & \text{转动关节} \\ {}^{i+1}R^i \dot{\omega}_i & \text{移动关节} \end{cases}$$

$${}^{i+1}v_{i+1} = \begin{cases} {}^{i+1}R({}^i v_i + {}^i \omega_i \times {}^i r_{i+1}) & \text{转动关节} \\ {}^{i+1}R({}^i v_i + {}^i \omega_i \times {}^i r_{i+1}) + \dot{d}_{i+1} {}^{i+1}z_{i+1} & \text{移动关节} \end{cases}$$

$${}^{i+1}\dot{v}_{i+1} = \begin{cases} {}^{i+1}R[{}^i \dot{v}_i + {}^i \dot{\omega}_i \times {}^i r_{i+1} + {}^i \omega_i \times ({}^i \omega_i \times {}^i r_{i+1})] & \text{转动关节} \\ {}^{i+1}R[{}^i \dot{v}_i + {}^i \dot{\omega}_i \times {}^i r_{i+1} + {}^i \omega_i \times ({}^i \omega_i \times {}^i r_{i+1})] + 2 {}^{i+1}\omega_{i+1} \times {}^{i+1}z_{i+1} \dot{d}_{i+1} + {}^{i+1}z_{i+1} \ddot{d}_{i+1} & \text{移动关节} \end{cases}$$

$${}^{i+1}\dot{v}_{Ci+1} = {}^{i+1}\dot{v}_{i+1} + {}^{i+1}\dot{\omega}_{i+1} \times {}^i r_{Ci+1} + {}^{i+1}\omega_{i+1} \times ({}^{i+1}\omega_{i+1} \times {}^i r_{Ci+1})$$

2 机械手动力学

- 计算关节力和力矩的内推公式 ($i = n \rightarrow 1$)

$${}^i f_{Ci} = m_i \dot{v}_{Ci}$$

$${}^i M_{Ci} = {}^i I_{Ci} {}^i \dot{\omega}_i + {}^i \omega_i \times {}^i I_{Ci} {}^i \omega_i$$

$${}^i f_i = {}^i f_{Ci} + {}_{i+1}^i R {}^{i+1} f_{i+1}$$

$${}^i M_i = {}^i M_{Ci} + {}_{i+1}^i R {}^{i+1} M_{i+1} + {}^i r_{i+1} \times {}_{i+1}^i R {}^{i+1} f_{i+1} + {}^i r_{Ci} \times {}^i f_{Ci}$$

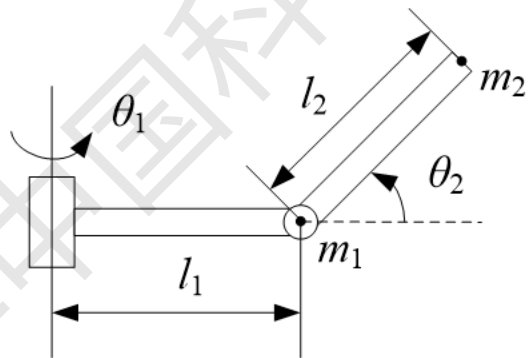
$$\tau_i = \begin{cases} {}^i M_i^T {}^i z_i & \text{转动关节} \\ {}^i f_i^T {}^i z_i & \text{移动关节} \end{cases}$$

自由运动时末端力为零

$${}^{n+1} f_{n+1} = {}^{n+1} M_{n+1} = 0$$

习题

- **习题1:** 简述串联关节机器人的动力学求解步骤。
- **习题2:** 如下图所示，两个连杆的质量均集中于连杆末端，连杆的长度分别为 l_1 和 l_2 ，其质量分别为 m_1 和 m_2 ，利用拉格朗日法建立该系统的动力学方程。



THANK YOU

